

# Technical Document #28: Retrieval Augmented Generation - Comprehensive Overview

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Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

### Key Takeaways:

- Semantic search finds documents based on meaning rather than exact keywords.
- Hallucination detection identifies when LLMs generate factually incorrect information.
- Query decomposition breaks complex queries into simpler sub-queries.